Experiment 10: 5G Network Slicing Environment Using MATLAB

Objectives, MATLAB Programs and Output Images

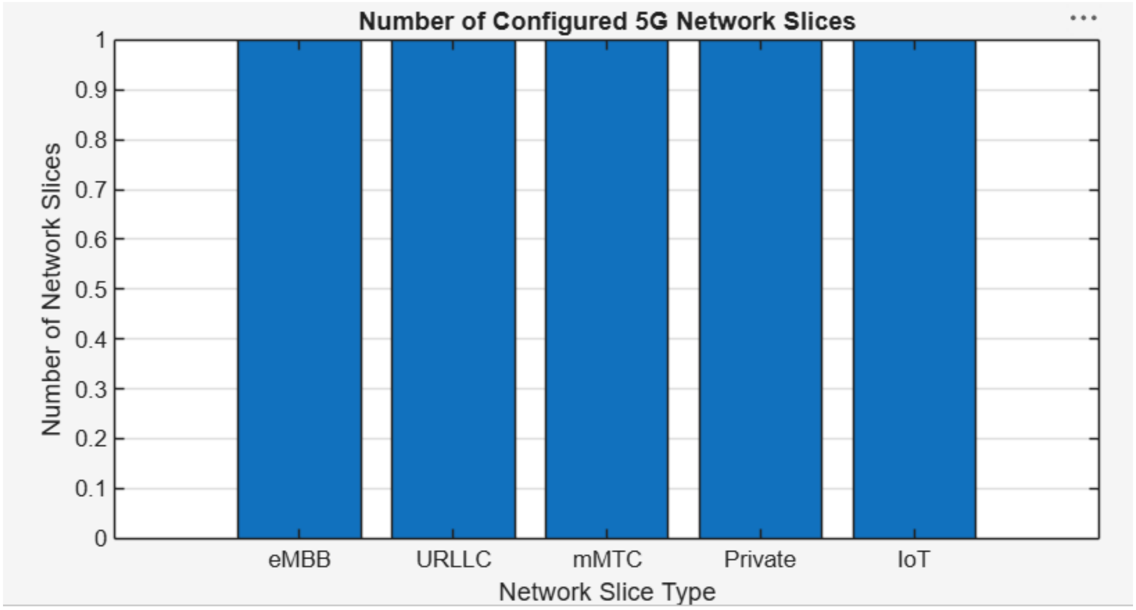
Objective 1

Analyze the Number of Network Slices Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB. Observe how different slices can be configured to support diverse service requirements and applications.

MATLAB Program

```
clc;
clear;
close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};

count = [1 1 1 1 1];       % One instance of each network slice

figure
bar(count)

grid on
set(gca,'XTickLabel',slices)

xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('Number of Configured 5G Network Slices')
```

Output Image – Objective 1



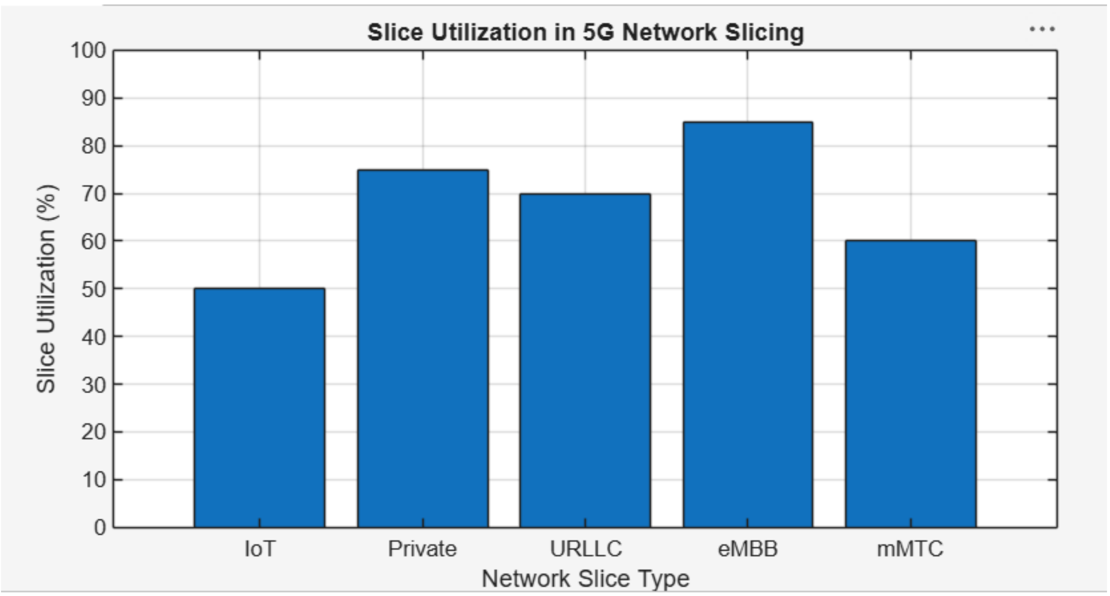
Objective 2

Compare Slice Utilization (%) Compare the utilization of different network slices using MATLAB. Analyze how efficiently network resources are allocated and utilized across multiple slices.

MATLAB Program

```
clc;
clear;
close all;

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
utilization = [85 70 60 75 50];

figure;
bar(slices, utilization);

grid on;
xlabel('Network Slice Type');
ylabel('Slice Utilization (%)');
title('Slice Utilization in 5G Network Slicing');
ylim([0 100]);
```

Output Image – Objective 2



Objective 3

Analyze End-to-End Latency (ms) Analyze the end-to-end latency of different network slices using MATLAB. Compare the latency values to understand the Quality of Service (QoS) requirements of various 5G applications.

MATLAB Program

```
clc;
clear;
close all;

slices = [1 2 3 4 5];
latency = [25 5 40 15 60];

figure;
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);

grid on;

xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});

xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title('5G Network Slicing: End-to-End Latency');

ylim([0 70]);
```

Output Image – Objective 3